

Mini Project: Exploratory Storytelling With Data

Project: The Sleep Debt – A Public Health Analysis

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Dataset: Sleep Health and Lifestyle Dataset

Data Source : Kaggle

Technical Insights: Sleep Health & Lifestyle Analysis

1. Problem Statement

In high-performance professional environments, sleep is often sacrificed in pursuit of productivity. This raises an important question: is this trade-off sustainable?

This project analyzes the Sleep Health and Lifestyle dataset to explore:

- Whether sleep quality improves consistently with longer sleep duration
- How stress levels vary across occupations
- Whether lifestyle and health factors influence sleep outcomes

The objective is to determine whether occupational stress contributes to measurable sleep disruption and potential health risk patterns.

2. Data Cleaning & Preprocessing

To ensure analytical accuracy and clean visualizations, the following preprocessing steps were performed:

- ★ **Handling Null Values:** The Sleep Disorder column contained NaN values representing individuals without diagnosed sleep conditions. These were replaced using `.fillna('None')` for clarity.
- ★ **Feature Engineering:** The categories "Normal" and "Normal Weight" in the BMI Category column were consolidated to remove redundancy and improve interpretability in the scatter plot.

3. Design Choices (Applying Visualization Principles)

To adhere to the "Remove Chartjunk" principle and improve clarity:

- ★ **Minimalist Aesthetics:** Applied `sns.despine()` to remove unnecessary borders and maintain a clean layout.
- ★ **Strategic Color Use:** Neutral grey was used for most professions, while Healthcare roles were highlighted in red (`#e74c3c`) to emphasize elevated stress levels.
- ★ **Sequential Palette:** Used in the sleep quality trend chart to reflect natural progression.

★ **Typography:** Increased font sizes for titles and labels to enhance readability.

3. Narrative Insight Summary

The 7-Hour Threshold

Sleep quality increases noticeably as individuals approach approximately seven hours of sleep. Beyond eight hours, improvements begin to plateau, suggesting the presence of an optimal recovery window.

Occupational Stress Differences

Healthcare professionals report the highest average stress levels compared to other occupations. This pattern suggests that professional environment may play a substantial role in sleep health outcomes.